



Eddy Current measuring and rail milling of defects: A case study from Finland

Carsten Jörn Rasmussen – Rail Maintenance Specialist
Linsinger

B O S T O N



Carsten Jörn Rasmussen



Rail Maintenance Specialist (MSE, IWE)

- **12 years** testing of rails at FORCE Technology, DK
- **12 years** as Engineering Rail Manager, Banedanmark
- **5 years** as Rail Maintenance Consult at COWI, DK
- **Today** – Rail Maintenance Specialist at Linsinger Maschinenbau, Austria

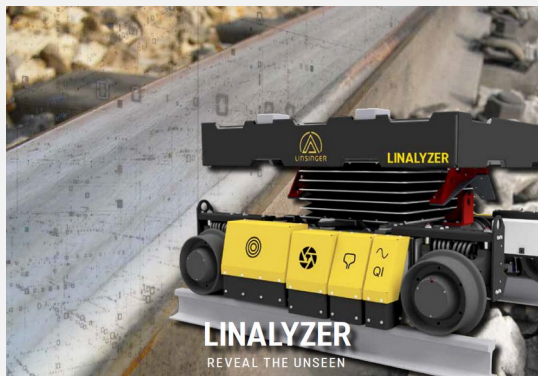




Eddy Current measuring and rail milling of defects



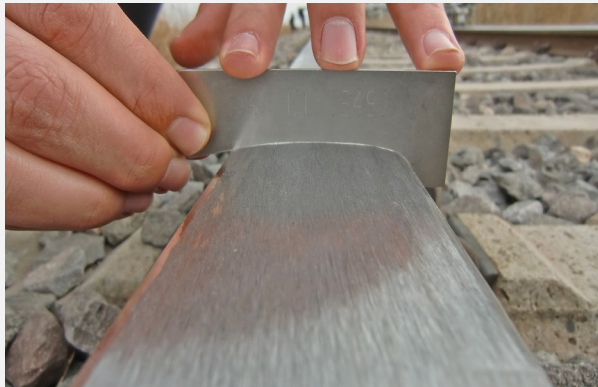
- Introduction to rail milling
- Case study from Finland
- Strategies of rail maintenance





Introduction to rail milling

- Mobile cutting process
- Clean and spark free
- Efficient removal of defects
- High rail surface quality
- Re-cycling of removed steel

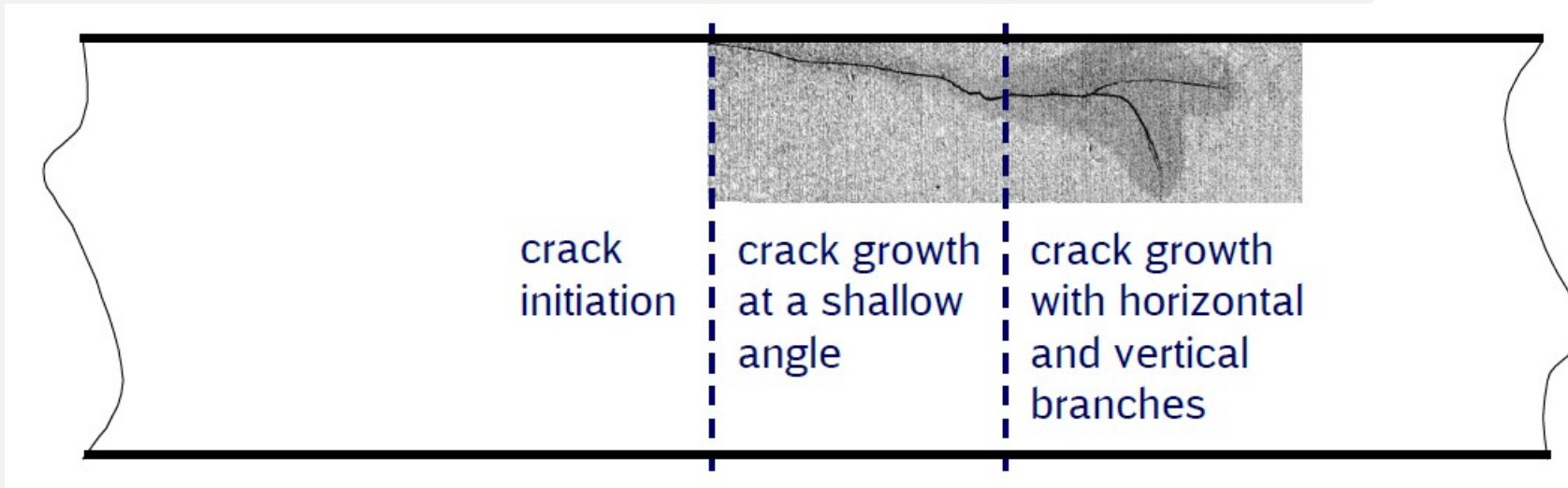




Why data-driven rail milling?



Efficient monitoring of rail surface
Efficient Removal of surface defects
in right time and location.



Ref. crack propagation according Dr. René Heyder, DB, Chengdu, CM2012



How does data-driven rail milling work?



Reprofiling process

1. Measure where and how much to remove

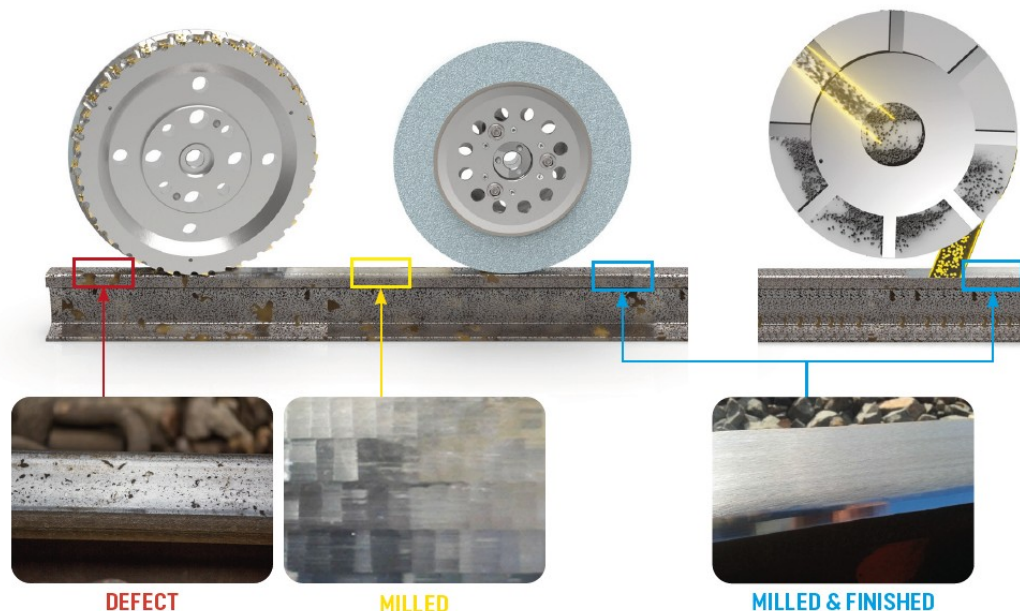


2. Milling

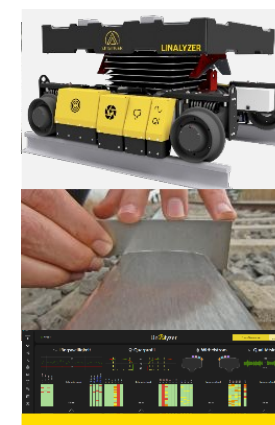
3. Finishing

- Polishing or

- LINSINGER Smart Surface Technology (LSST)

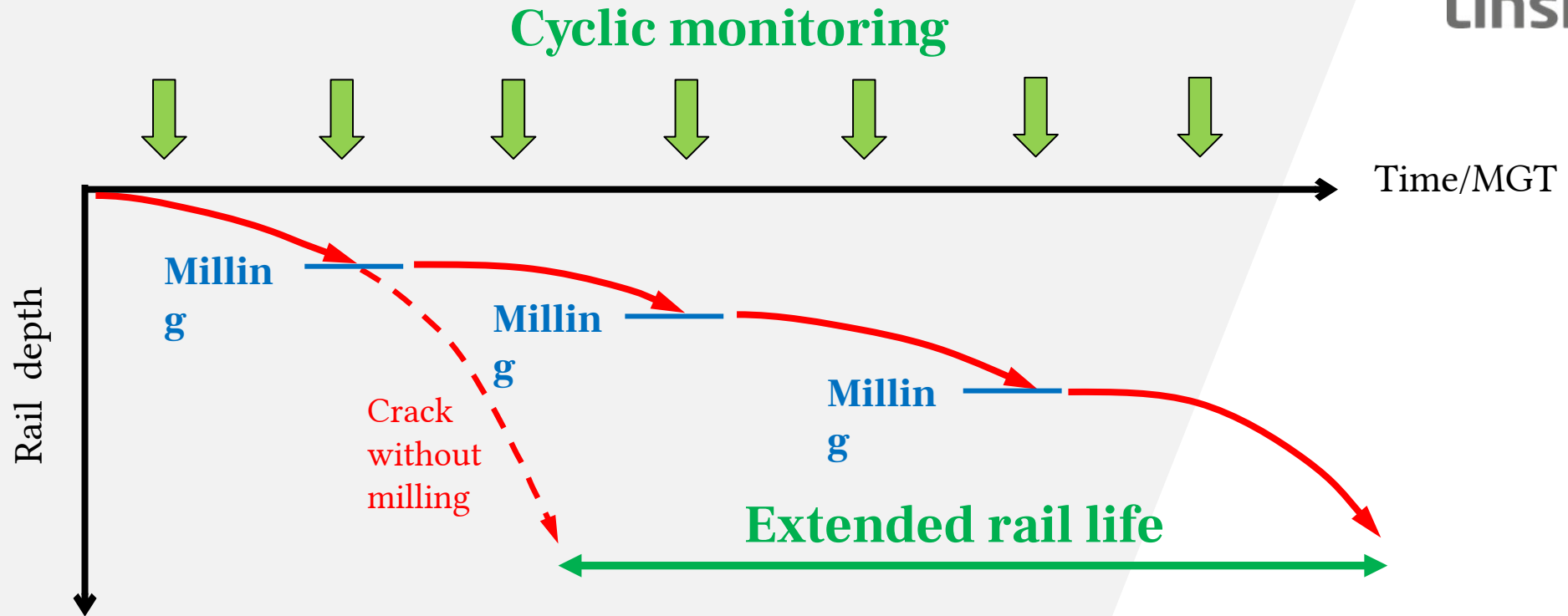


4. Documentation of a reprofiled and defect free rail





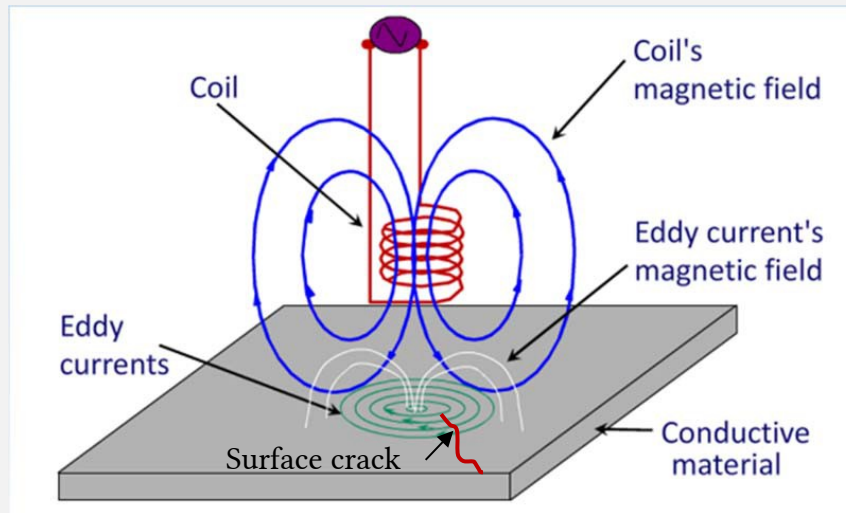
Data-driven rail milling strategy



Data-driven milling, but ONLY when and where it is needed with efficient removal of the defects!



Eddy Current Testing (ET)



Ref: Gordon Bachinsky, WRI, Transit, 2017

- The probe's coil creates the primary magnetic field.
- Eddy currents form in the test material and generate their own secondary magnetic field.
- Cracks near the surface disrupt the flow of these eddy currents and the secondary magnetic field.
- The changes in the secondary magnetic field interact with the primary magnetic field of the coil.
- These changes are displayed on an instrument and interpreted to detect cracks.

Measuring of surface cracks with depth 0-5 mm.



Walking or driving

5 km/h to 40 km/h

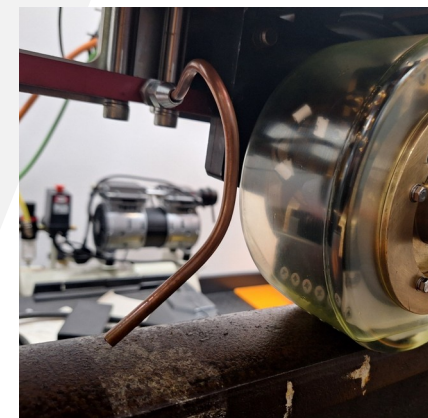
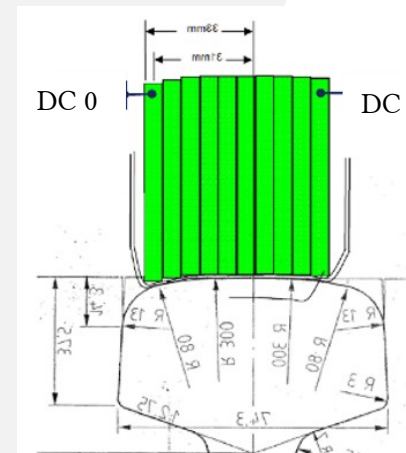
Gauge corner



B O S T O



Gauge corner ♦ top





Case study from rail milling by Linmag in Finland in 2025





Why did the customer in Finland call Linmag?



Rail Conditions:

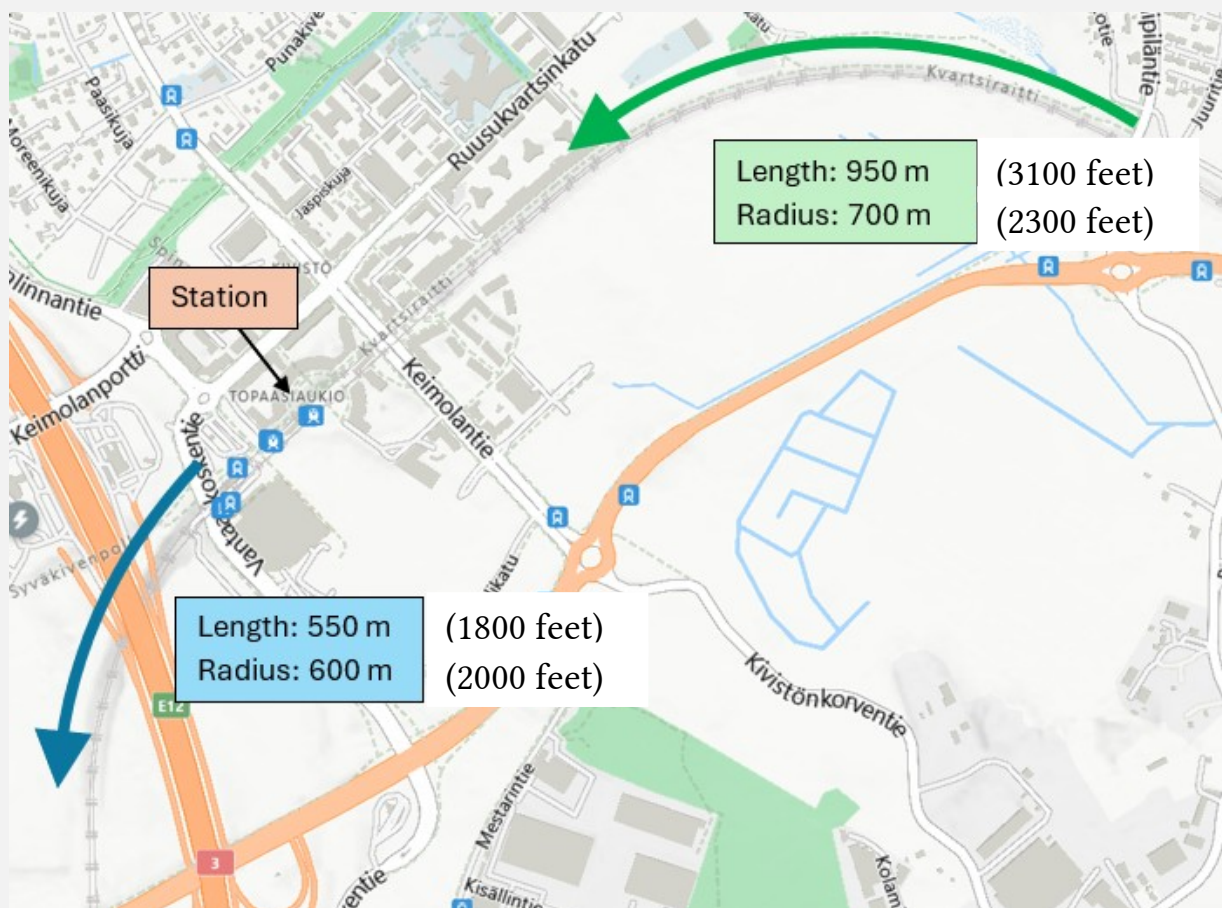
- RCF damage in rails
- Mainly head checks
- Some corrugation
- Costly rail exchanges
- New target rail profile



[illegible]



Case study: Rail milling in two curves



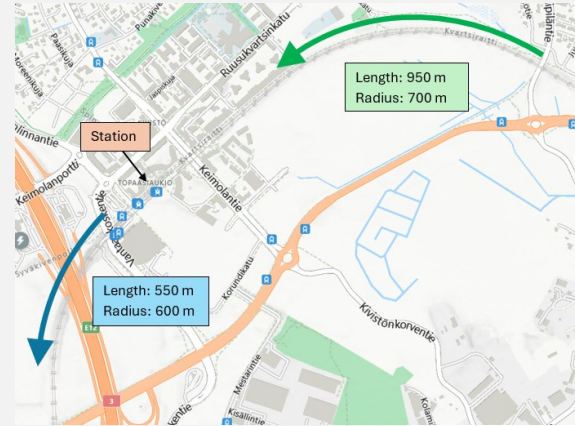
head check cracks in
outer rails in curves



Case study: Rail milling in two curves

Rails in the two curves:

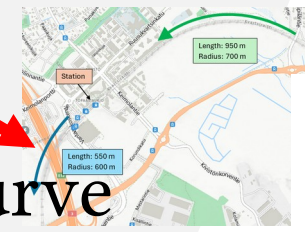
- E60E1, R260, 1:40
- Rails 10 years in service
- Total traffic load is 70 MGT
- Vertical rail wear limit 14 mm
- Curve radius 600/700 m





Case 1: Data-driven milling in high rail in 550 m curve

(1800 feet)



(0.08 – 0.12 In)

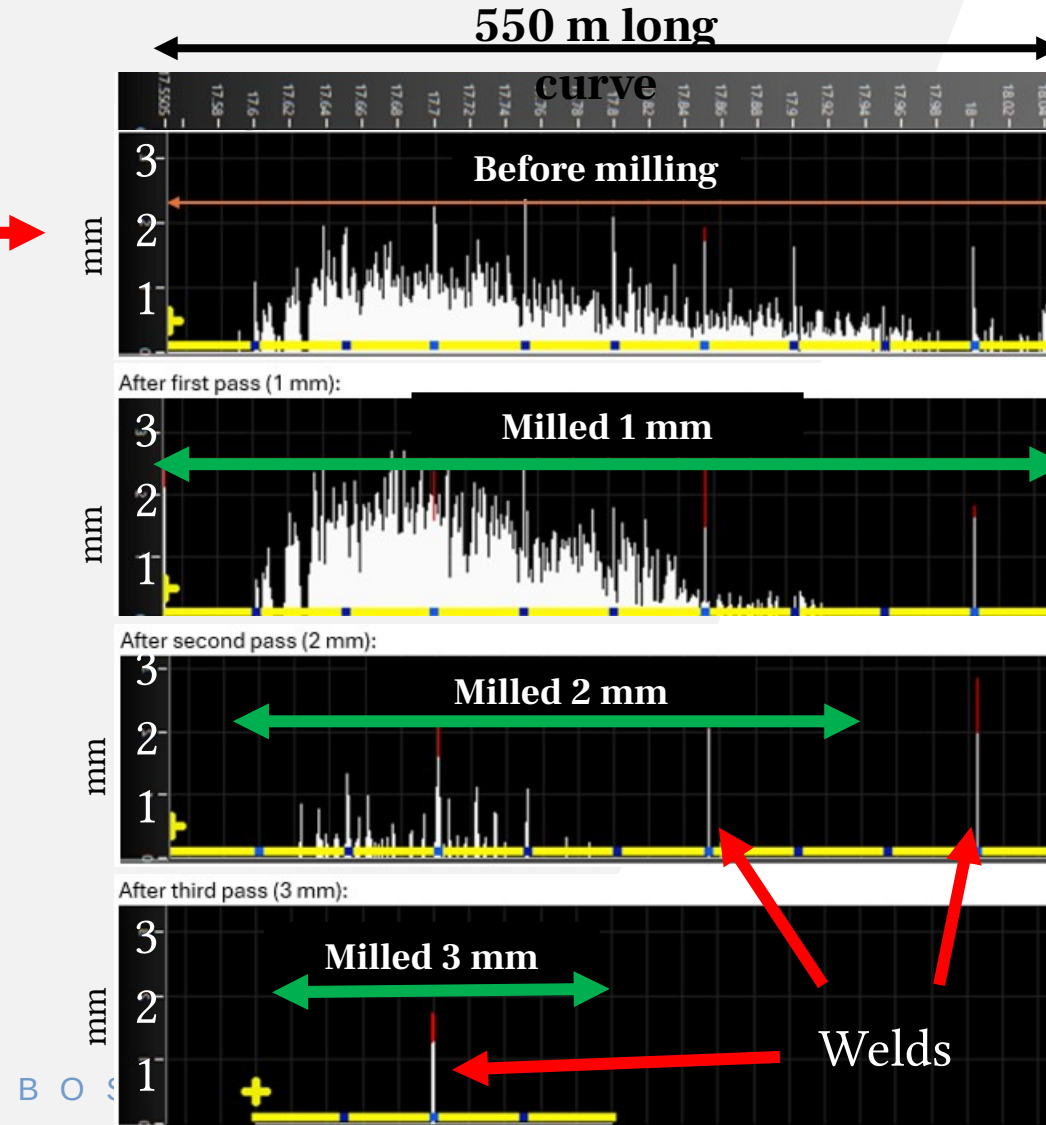
2-3 mm deep
head checks



Defect free
rails in all the
550 m curve
after milling
of up to 3 mm



Only milling where
needed thanks to Eddy
Current!



Eddy current
measurement
s of crack
depth

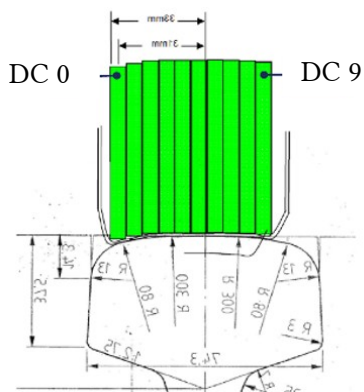
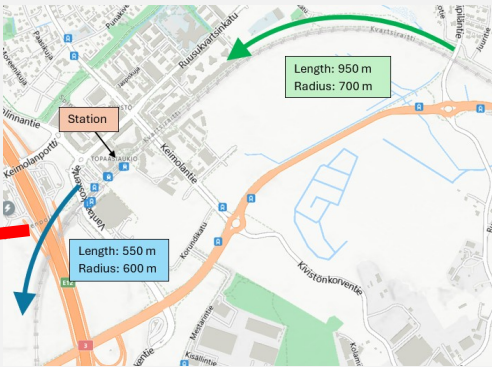
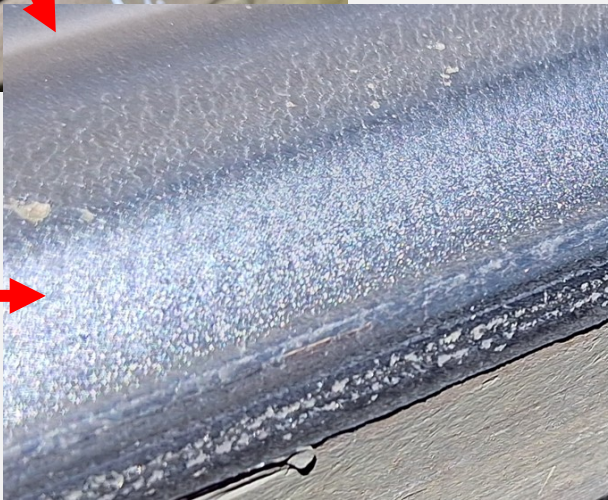
Case 1: Head Checks in km 17.709



Km 17.709



Head check cracks.
Hard to see visually!



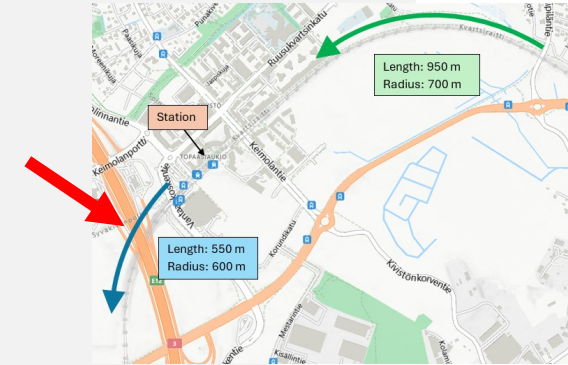
Eddy Current measuring of up to 2,5 mm deep head checks cracks in the high rail in km 17.709.

17.708	0	0.2	0.9	0	0	0	0	0	0	0	0
17.708	0.2	0.3	0.4	0	0	0	0	0	0	0	0
17.708	0.5	0.2	0.9	0	0	0	0	0	0	0	0
17.708	0	0.4	0.9	0	0	0	0	0	0	0	0
17.708	0.4	0.7	1.6	0	0	0	0	0	0	0	0
17.708	0	0.2	0.3	0	0	0	0	0	0	0	0
17.708	0	0	0.2	0	0	0	0	0	0	0	0
17.708	0	0.5	1.6	0	0	0	0	0	0	0	0
17.708	0.2	0.5	0.6	0	0	0	0	0	0	0	0
17.708	0	0.4	1.1	0	0	0	0	0	0	0	0
17.709	0	0.5	0.7	0	0	0	0	0	0	0	0
17.709	0.4	0.6	1.6	0.2	0	0	0	0	0	0	0
17.709	0.4	0.3	1.4	0	0	0	0	0	0	0	0
17.709	0	0.4	1.1	0	0	0	0	0	0	0	0
17.709	0	0.8	2.3	0	0	0	0	0	0	0	0
17.709	0	0.4	1.3	0	0	0	0	0	0	0	0
17.709	0.2	0.3	1.8	0	0	0	0	0	0	0	0
17.709	0	0.2	0.6	0	0	0	0	0	0	0	0
17.709	0	0	0	0	0	0	0	0	0	0	0
17.709	0	0	0.2	0	0	0	0	0	0	0	0
17.710	0	0.7	2.3	0	0	0	0	0	0	0	0
17.710	0.2	0.6	2.4	0.3	0	0	0	0	0	0	0
17.710	0.5	0.3	0.7	0	0	0	0	0	0	0	0
17.710	0	0.8	2.5	0.2	0	0	0	0	0	0	0
17.710	0.3	0.3	1.3	0	0	0	0	0	0	0	0
17.710	0.4	0.4	1.2	0	0	0	0	0	0	0	0
17.710	0	0	0	0	0	0	0	0	0	0	0
17.710	0	0	0	0	0	0	0	0	0	0	0
17.710	0	0	0	0	0	0	0	0	0	0	0
17.710	0	0.3	0.6	0	0	0	0	0	0	0	0

(10 feet)
3 m



Case 2: RCF, field side, low rail, km 17.825



(0.08 In)
**2 mm deep
 cracks to the
 field site in the
 low rail**



Remaining cracks of up to approx. 1 mm in low rail to the field site.
(0.04 In.)



17.822	0	0	0	0	0	0	0	0	0
17.823	0	0	0	0	0	0	0	0.4	0.2
17.823	0	0	0	0	0	0	0	0.6	0.4
17.823	0	0	0	0	0	0	0	0.2	0.4
17.823	0	0	0	0	0	0	0	0.2	0.3
17.823	0	0	0	0	0	0	0	0.3	0.2
17.823	0	0	0	0	0	0.2	0.2	0.4	0.5
17.823	0	0	0	0	0	0	0.2	0.6	0.7
17.823	0	0	0	0	0	0	0	0.5	0.4
17.823	0	0	0	0	0	0	0	0.5	0.5
17.823	0	0	0	0	0	0	0.4	0.8	0.5
17.823	0	0	0	0	0	0	0.3	0.8	0.4
17.824	0	0	0	0	0	0.3	0.3	0.8	0.6
17.824	0	0	0	0	0	0.6	0.5	0.9	0.7
17.824	0	0	0	0	0	0.3	0	0.7	0.6
17.824	0	0	0	0	0	0	0	0.5	0.4
17.824	0	0	0	0	0	0.4	0.3	0.6	0.6
17.824	0	0	0	0	0	0.2	0	1.1	0.4
17.824	0	0	0	0	0	0	0	0.4	0.3
17.824	0	0	0	0	0	0	0	0.4	0.5
17.824	0	0	0	0	0	0	0	0.7	0.5
17.824	0	0	0	0	0	0	0	0.4	0.6
17.825	0	0	0	0	0	0.3	0.3	0.5	0.6
17.825	0	0	0	0	0	0	0	1	0.5
17.825	0	0	0	0	0	0	0	0.5	0.3
17.825	0	0	0	0	0	0	0.2	0.6	0.3
17.825	0	0	0	0	0	0	0.3	0.7	0.5
17.825	0	0	0	0	0	0	0	0.4	0.3
17.825	0	0	0	0	0	0	0	0.5	0.4
17.825	0	0	0	0	0	0.2	0	0.7	0.4
17.825	0	0	0	0	0	0	0.3	0.9	0.5
17.825	0	0	0	0	0	0	0.3	1.2	0.6
17.826	0	0	0	0	0	0	0	0.3	0.3
17.826	0	0	0	0	0	0	0	0	0
17.826	0	0	0	0	0	0	0.2	0.5	0.3
17.826	0	0	0	0	0	0	0	0.4	0.3
17.826	0	0	0	0	0	0	0	0.5	0.3
17.826	0	0	0	0	0	0	0	0.3	0.3
17.826	0	0	0	0	0	0	0	0.3	0
17.826	0	0	0	0	0	0	0	0.2	0.2
17.826	0	0	0	0	0	0	0	0.3	0.3
17.826	0	0	0	0	0	0	0	0.4	0.4
17.827	0	0	0	0	0	0	0	0	0.3
17.827	0	0	0	0	0	0	0	0.3	0
17.827	0	0	0	0	0	0	0	0	0

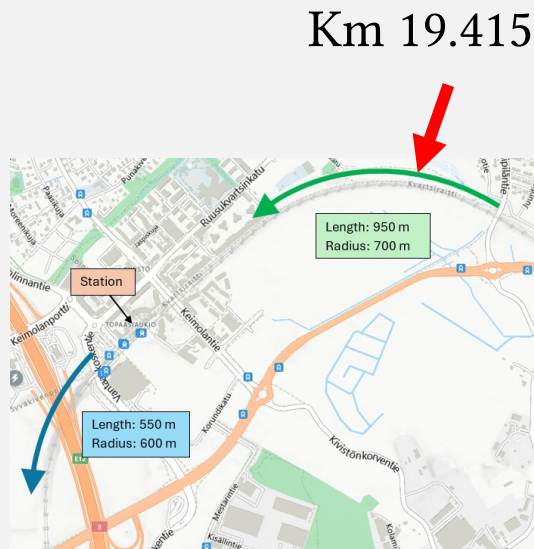
(14 feet)
4 m

1





Case 3: Head Checks in high rail km 19.415





Case 3: Head Checks in high rail km 19.415



LINSINGER

1 mm removed.

**More severe
spallation**





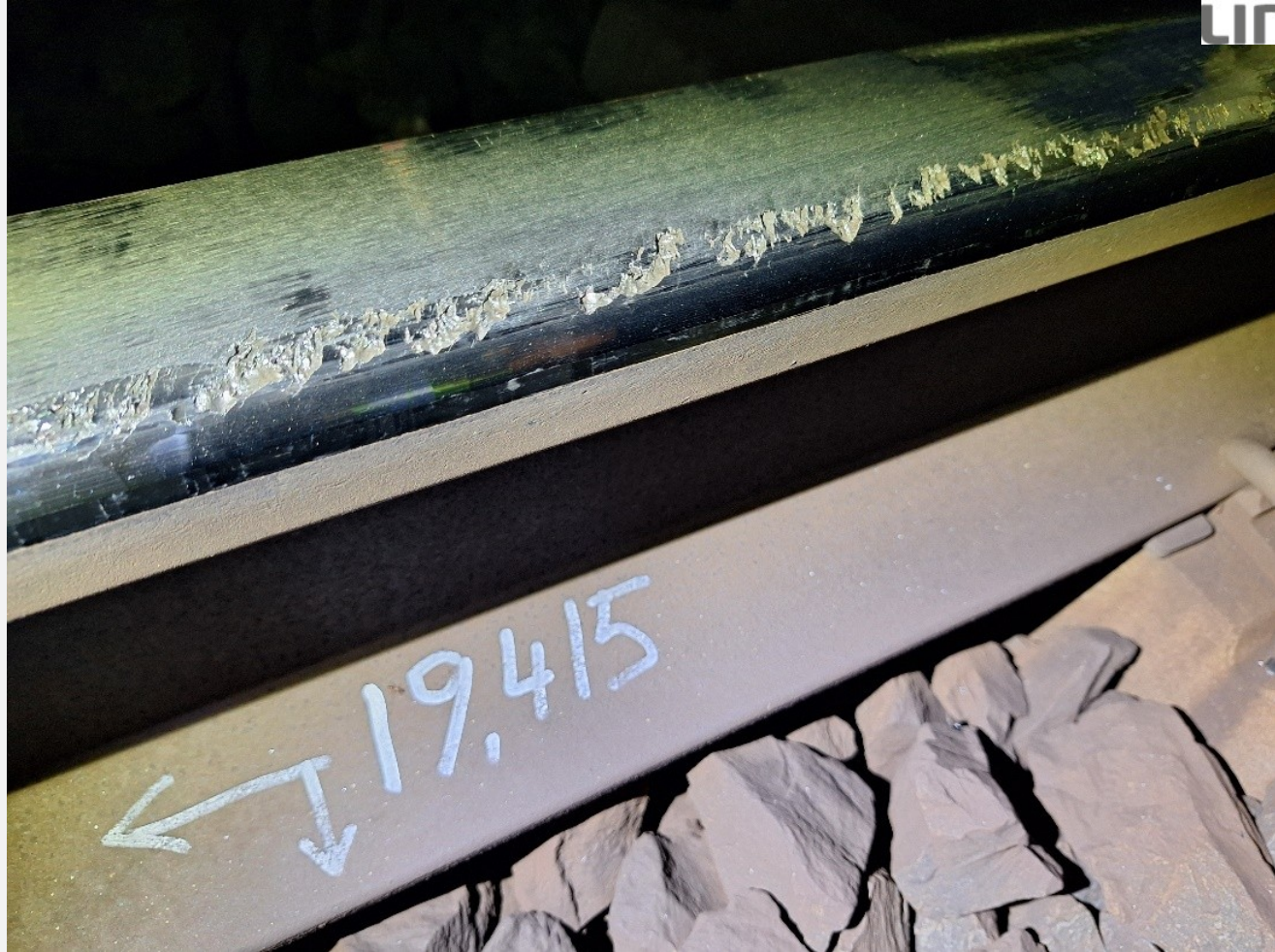
Case 3: Head Checks in high rail km 19.415



LINSINGER

2 mm removed.

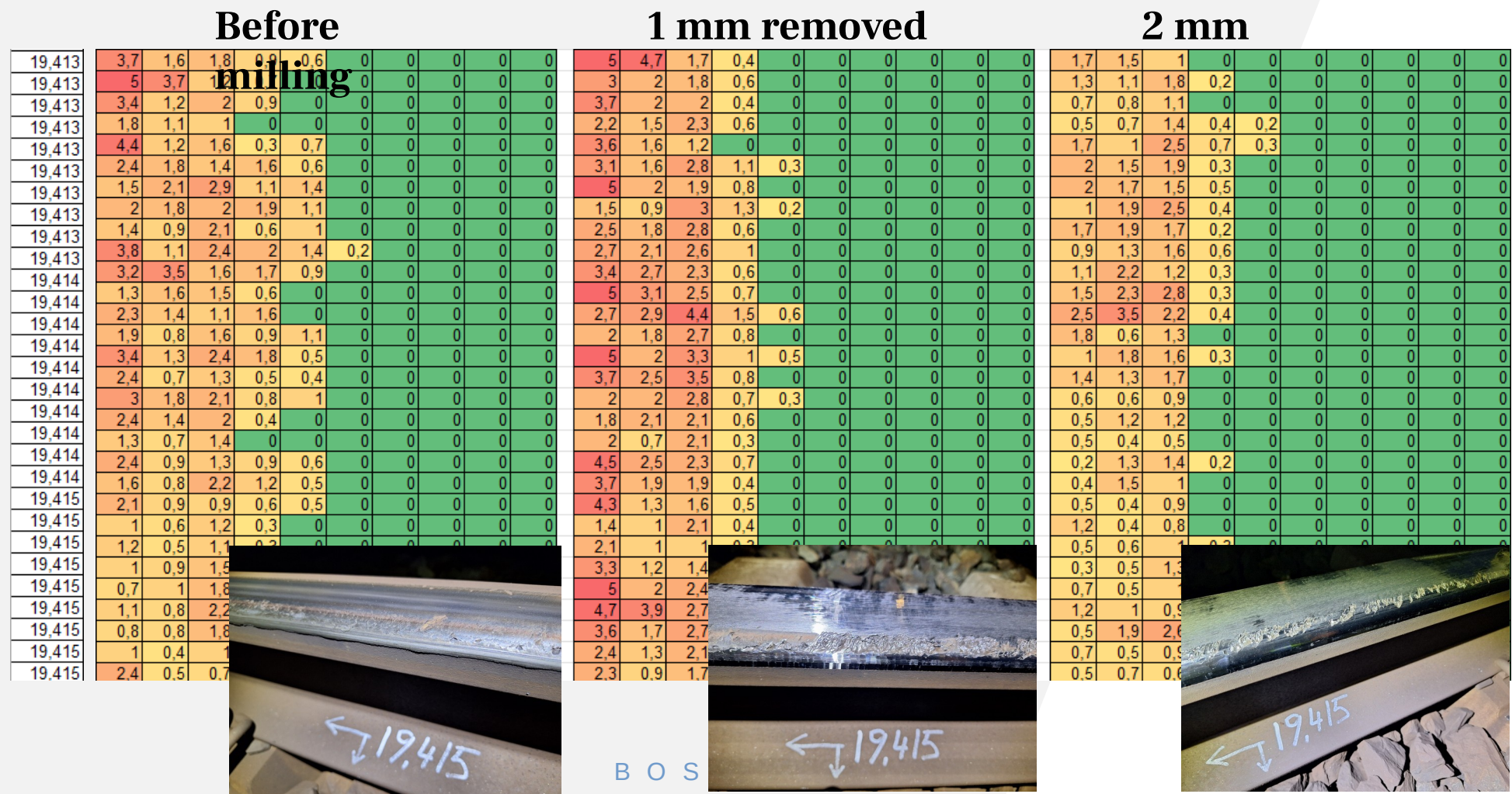
**Spallation and
damages
reduced**





Case 3: Head Checks in high rail km 19.415

Eddy current measurements





Case 3: Head Checks in high rail km 19.415



3 mm removed

Looks better



4 mm removed

Looks defect free



5 mm removed

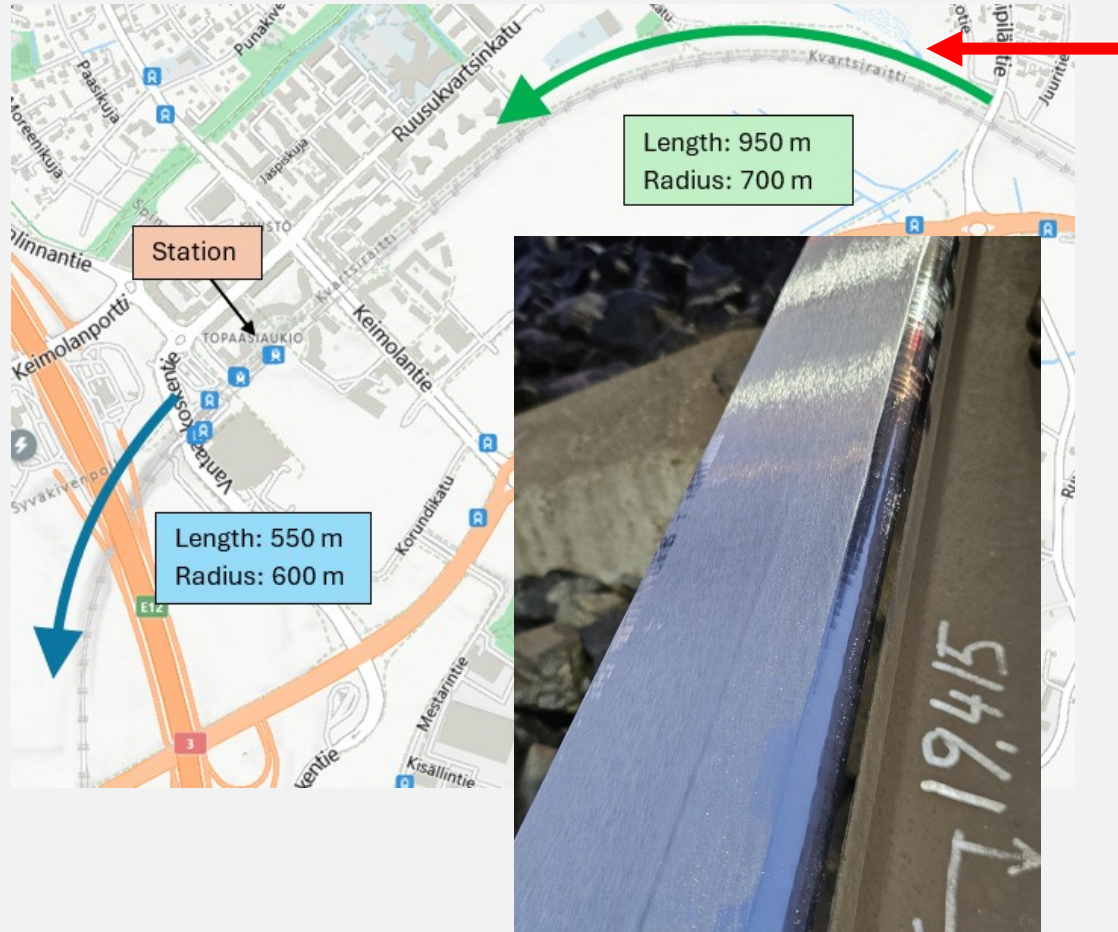
All defects







Extended rail service life-time by milling in green curve



4-5 mm deep
Head Checks
after 10 years
and 70 MGT

- Natural rail wear only 1.5 mm
- Removal up to 5.0 mm
- 6,5 mm and wear limit 14 mm

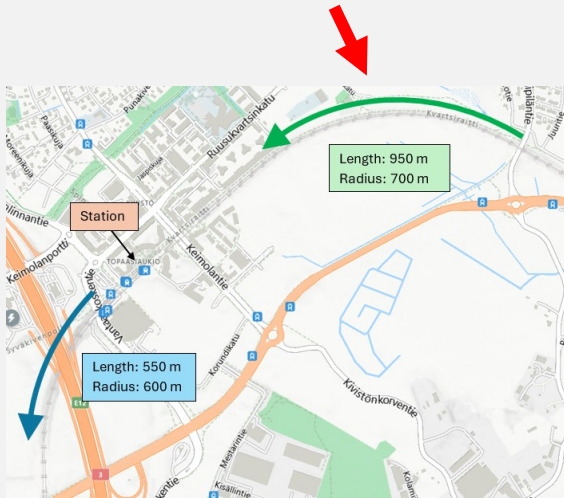
**The milling has at least
doubled the rails
lifetime**



Rail maintenance strategy in green curve



Severe 5 mm deep damages in green curve after 10 years and total 70 MGT



Would it be possible to extent the rails lifetime even further by following a more optimal reprofiling strategy from rail





Strategies of rail maintenance

Data-driven Milling



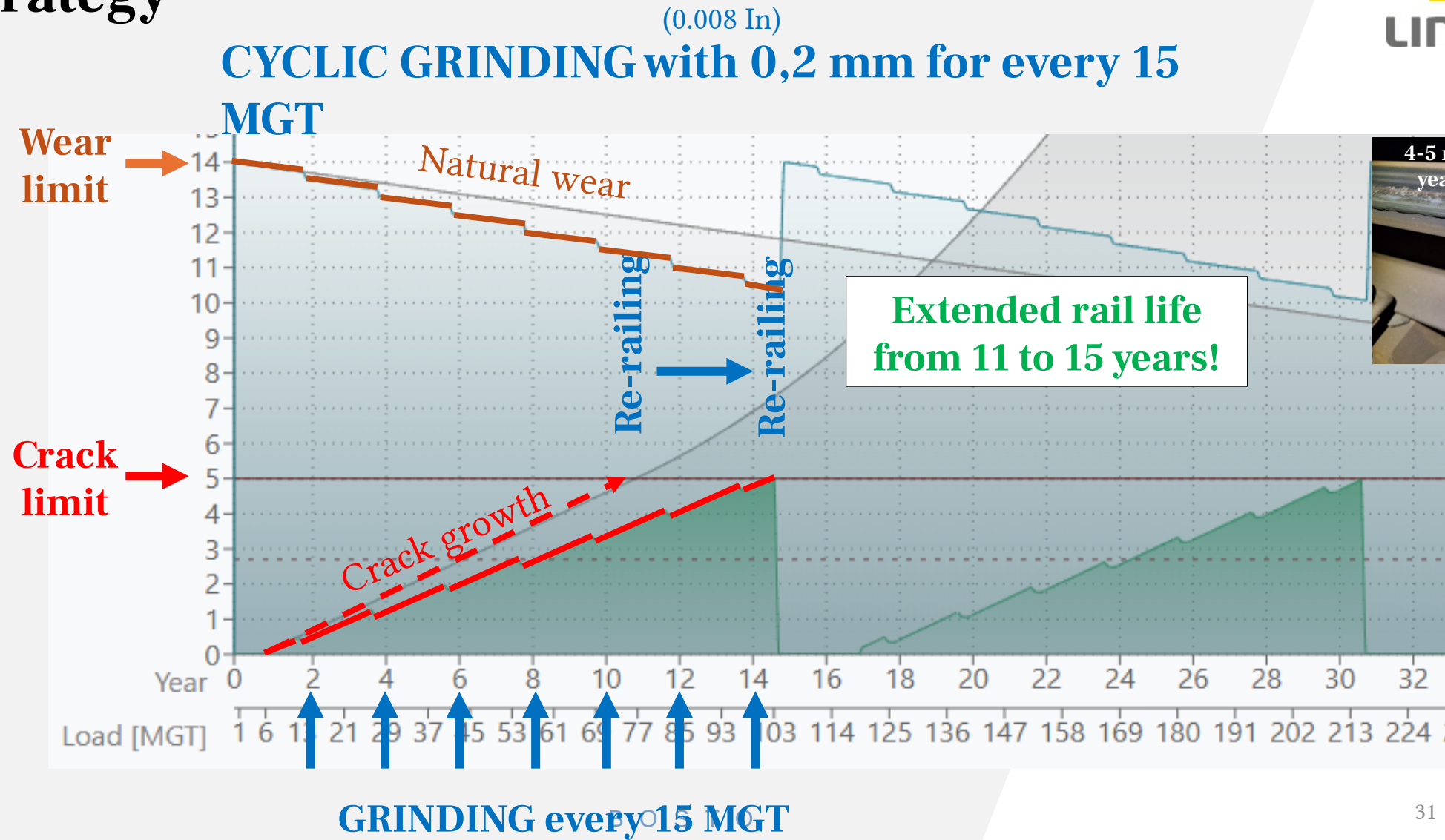
Cyclic Grinding



Estimation of rail life-time by a **cyclic grinding** strategy



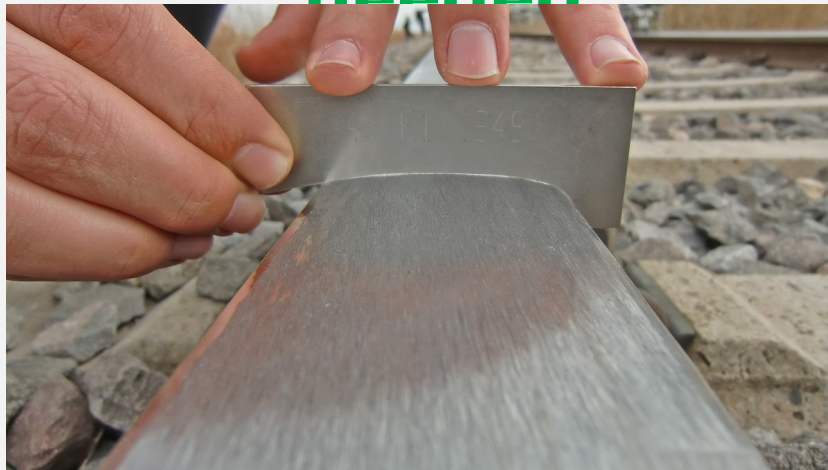
Case





Recommended rail maintenance strategy

- ✓ **Efficient Monitoring**
- ✓ **Data-driven Rail Milling**
- ✓ **Maintenance When, Where and How as needed**



Eddy Current Testing and Rail Milling of Defects in Finland

**Thank You for
Your Attention**



Contacts:

Elvis Kozica, Senior Sales Manager, C. +43 7613 8840-0, e.kozica@linsinger.com

Carsten Jörn Rasmussen, Rail Maintenance Specialist, C. +43 699 188 40 058, c.rasmussen@linsinger.com

